

Git Interview Questions

A set of 20 questions spanning basic to advanced Git concepts

Basics

1. What is Git, and how does it differ from a centralized version control system like SVN?
2. What is the difference between 'git fetch' and 'git pull'?
3. Explain the difference between 'git merge' and 'git rebase'.
4. What is a merge conflict, and how would you resolve one?
5. What does 'git clone' do, and how is it different from 'git init'?

Staging, Commits & History

1. What is the staging area (index) in Git, and why does it exist?
2. How do you undo a commit that hasn't been pushed yet?
3. What is the difference between 'git reset', 'git revert', and 'git checkout'?
4. How would you squash multiple commits into one?
5. What does 'git cherry-pick' do, and when would you use it?

Branching & Collaboration

1. What is a branch in Git, and how is it implemented internally?
2. Explain a common branching strategy (e.g., Git Flow or trunk-based development).
3. How do you resolve a conflict during a rebase?
4. What is a fast-forward merge versus a three-way merge?
5. How do you delete a remote branch, and how is that different from deleting a local branch?

Advanced & Internals

1. What are Git hooks, and can you give an example use case?
2. Explain how Git stores data internally (blobs, trees, commits).
3. What is 'git stash', and how would you use it in a typical workflow?
4. What is the difference between 'git rebase -i' and a regular rebase?
5. What is a detached HEAD state, and how do you recover from it?